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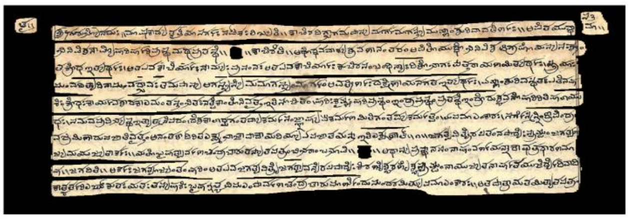
1. Problem Description

Pal Leaf Denoising/ Region of Interest extraction – In this problem we denoise a manuscript image. These manuscripts are ages older than us, ofcourse they are degraded and damaged. The core idea of this problem is we extract some regions of interest (like figures/ painting/ text) from these manuscript images.

Here the input is a palm leaf manuscript image and the output is a text region only extracted from that palm leaf image.

Example:

Input



Output

2. Theory

2.1 Swin Transformer Block: The core building block is the Swin Transformer Block, which replaces global self-attention ($O(n^2)$) with Window based Multi head Self-Attention (W-MSA) computed within local M×M windows, and Shifted Window MSA (SW-MSA) in alternating layers to allow cross-window information flow, reducing the complexity to O(N). Fig.2 of [1] shows the block working.

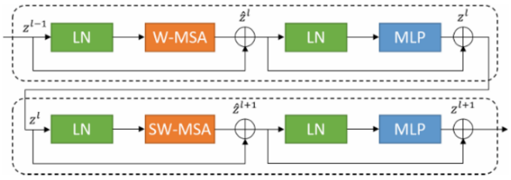


Fig. 2. Swin transformer block.

2.2 Swin-Unet Architecture: Swin-Unet is a pure Transformer U-Net with no convolutions. It follows the classic encoder–bottleneck–decoder structure with skip connections, but replaces all CNN components with Swin Transformer blocks.

Component	Operation	Resolution Change
Patch Partition	Split 224×224 into 4×4 patches → 56×56 tokens	÷4
Linear Embedding	Project 48-dim patches to C=96 dims	—
Encoder (×3)	Swin Blocks + Patch Merging (2× downsample)	÷2 per stage
Bottleneck	2× Swin Blocks (no size change)	—
Decoder (×3)	Patch Expanding (2× upsample) + Swin Blocks	×2 per stage
Final Expand	4× Patch Expanding to full resolution	×4
Linear Projection	Map C-dim features → num_classes (2)	—

Fig. Architecture Parameters

3. Dataset Used

Malayalam is among the 22 scheduled languages and 11 classical languages of India, with over 35 million speakers.

LeafOCR-Line, a Malayalam palm leaf manuscripts dataset from 20 different palm leaf bundles consisting of 1710 text line masked manuscripts with corresponding deterioration levels for each manuscript.

Table 3 from [2] shows the metadata of the dataset.

Metadata item	Description
Dataset Name	LeafOCR-Line
Number of Images	1710 images from 20 different palm leaf bundles
Average Text Lines	More than 10000 text lines can be extracted from the images using their corresponding masks.
Deterioration Levels	A total of 405 images are categorized as more deteriorated, 394 images as less deteriorated, and 911 images as medium deteriorated
Data Resource	Shijualex repository
Annotation	Manually using CVAT
Applications	Palm leaf digitization

Table 3. Dataset Metadata for LeafOCR-Line.

4. Model Used – SWIN-UNET

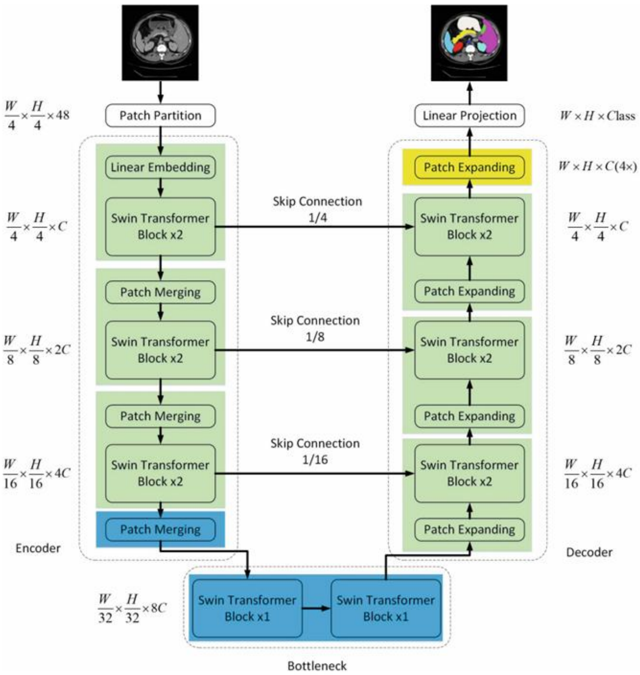


Fig. 1. The architecture of Swin-Unet, which is composed of encoder, bottleneck, decoder and skip connections. Encoder, bottleneck and decoder are all constructed based on Swin transformer block.

Fig.1([1]) shows the architecture of the used model.

The following table shows the hyperparameters used during training.

Hyperparameter	Value	Hyperparameter	Value
Input Size	224 × 224	Embed Dim (C)	96
Patch Size	4 × 4	Encoder Depths	[2, 2, 2, 2]
Window Size	7 × 7	Decoder Depths	[1, 2, 2, 2]
Num Classes	2 (bg/text)	Num Heads	[3, 6, 12, 24]
Parameters	~27M	Upsampling	Patch Expanding
Optimizer	SGD (mom=0.9)	Loss	0.5×Dice + 0.5×CE
Base LR	0.01	Warmup	10 epochs (of 100)
AMP	Disabled (CPU)	Batch Size	2 (CPU constraint)

5. Pipeline followed

1 Data Preparation:

Leaf-OCR dataset loaded.
Ground-truth masks read as RGB, any non-black pixel = text (label 1).
Stem-name matching pairs images to masks.
Train/val/test splits performed.

2 Augmentation (Train):

Random Resized Crop(224×224), Shift Scale Rotate (±3°), Elastic Transform, Random Brightness Contrast / CLAHE / Hue Saturation Value, Gauss Noise, Coarse Dropout, ImageNet normalization (mean/std).

3 Model Initialisation:

Swin-Unet Palm Leaf wrapper instantiated.
Swin-Tiny pretrained Image Net weights loaded into encoder via load_from().
Head replaced with 2-class output. ~27M parameters total.

4 Training Loop:

SGD (lr=0.01, momentum=0.9, wd=1e-4) + Cosine Warmup LR (10 epochs).
Loss = 0.5×Dice + 0.5× Cross Entropy.
Best checkpoint saved by val DSC.
History JSON saved every epoch.

5 Variable-Size Inference:

At test time: pad image to stride-32 multiple → bilinear resize to 224×224 → Swin-Unet → bilinear upsample back to padded size → crop to original resolution.
Threshold probs ≥ 0.5 → binary mask.

6 Output Visualisation:

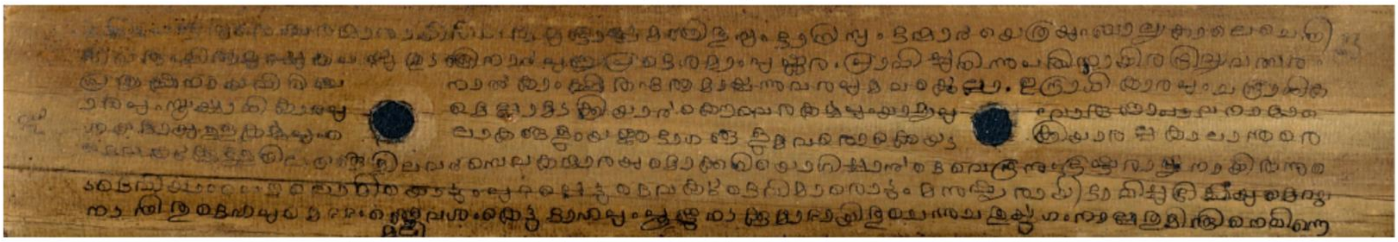
Predicted binary mask applied to original image:
pixels where pred=1 show the original manuscript pixels, background zeroed to black.

6. Results

The model was trained for 2 epoch on CPU, yielding
DiceCELoss= 0.7641,
Dice-Similarity coefficient (DSC) = 0.503794369766513 and
average Hausdorff Distance (HD): 1.0

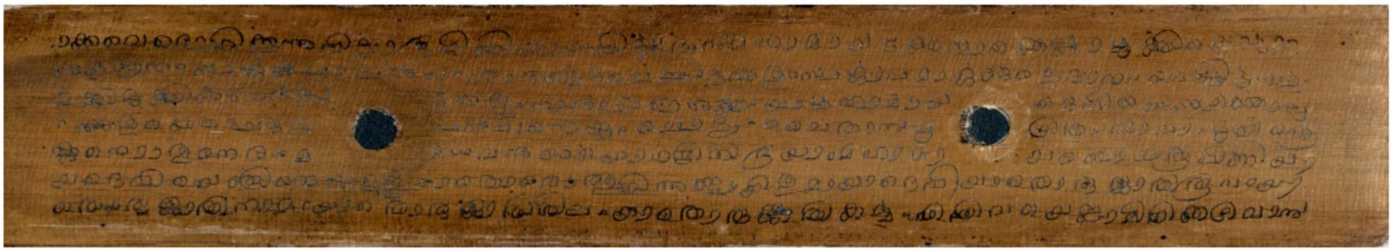
Sample input and outputs:

Input — 846_47

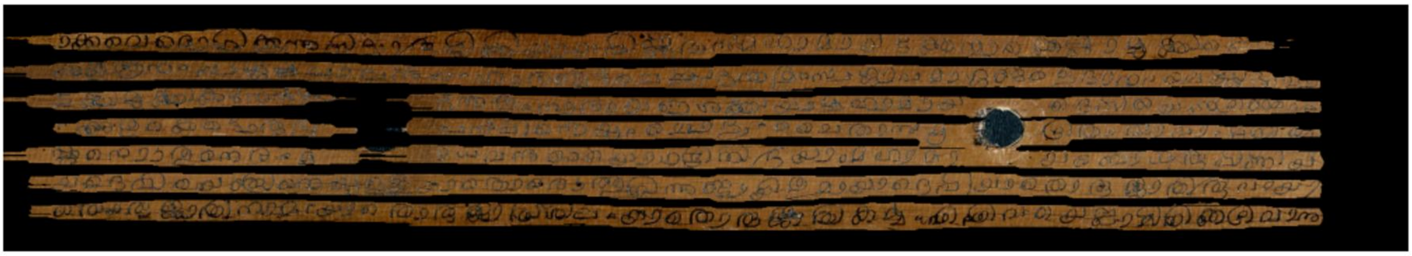


Output

Input — 846_12



Output



7. References

[1] H. e. Cao, "Swin-Unet: Unet-Like Pure Transformer for Medical Image Segmentation," in *Springer Nature*, Switzerland, 2023.

[2] R. Sivan and P. B. Pati, "A benchmark dataset for text line segmentation in palm leaf documents," *Sci Data,* 2026.